

Norm-One Periodic Decompositions for Generic Systems and for Noncommensurable Triples

Candidate partial-result packet for arXiv:1312.3798

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Status. Candidate strong partial result; likely valid, pending human review. The translation decomposition constant is one for every system with rationally independent reciprocal periods, and for every three-period system which is not wholly commensurable.

1 Source question

Farkas and Révész ask whether the optimal constant in a bounded continuous periodic decomposition can always be chosen to be one [1, problem following Remark 3.5]. More precisely, if

$$\mathcal{P}_\alpha = \{g \in BC(\mathbb{R}) : g(x + \alpha) = g(x)\}, \quad \alpha \neq 0,$$

and $f \in \mathcal{P}_{\alpha_1} + \cdots + \mathcal{P}_{\alpha_n}$, must there be $f_j \in \mathcal{P}_{\alpha_j}$ with

$$f = \sum_{j=1}^n f_j, \quad \max_j \|f_j\|_\infty \leq \|f\|_\infty?$$

The source proves existence of decompositions and records the sharp constant one for two periods. The exact source problem is reproduced in `source_excerpt.pdf`.

2 Results

Two nonzero reals are called *commensurable* when their ratio is rational.

Theorem 1 (Rationally independent reciprocal periods). *Suppose $\alpha_1^{-1}, \dots, \alpha_n^{-1}$ are linearly independent over $\mathbb{Q}$. Every $f \in \sum_{j=1}^n \mathcal{P}_{\alpha_j}$ has a decomposition $f = \sum_j g_j$, $g_j \in \mathcal{P}_{\alpha_j}$, such that*

$$\|g_j\|_\infty \leq \|f\|_\infty \quad (j = 1, \dots, n).$$

The constant one is sharp.

Theorem 2 (Three periods). *Suppose $\alpha_1, \alpha_2, \alpha_3$ are not all commensurable. Every $f \in \mathcal{P}_{\alpha_1} + \mathcal{P}_{\alpha_2} + \mathcal{P}_{\alpha_3}$ has a decomposition $f = g_1 + g_2 + g_3$, $g_j \in \mathcal{P}_{\alpha_j}$, with*

$$\max_{1 \leq j \leq 3} \|g_j\|_\infty \leq \|f\|_\infty.$$

Again the constant one is sharp.

The second theorem goes beyond Theorem 1: three pairwise incommensurable reciprocal periods can satisfy an integer relation, so their orbit closure can be a proper two-dimensional subtorus of $\mathbb{T}^3$.

3 The full-product argument

Proof of Theorem 1. Write $\beta_j = \alpha_j^{-1}$ and choose an initial decomposition

$$f(x) = \sum_{j=1}^n F_j(\beta_j x \bmod 1), \quad F_j \in C(\mathbb{T}, \mathbb{R}).$$

Kronecker density says that $x \mapsto (\beta_1 x, \dots, \beta_n x) \bmod 1$ is dense in $\mathbb{T}^n$. Put

$$a_j = \min F_j, \quad b_j = \max F_j, \quad c_j = \frac{a_j + b_j}{2}, \quad r_j = \frac{b_j - a_j}{2},$$

and $c = \sum_j c_j$, $R = \sum_j r_j$. Density gives

$$\min f = c - R, \quad \max f = c + R, \quad M := \|f\|_\infty = |c| + R.$$

There are real numbers d_j such that

$$\sum_j d_j = c, \quad |d_j| \leq M - r_j.$$

Indeed, the symmetric intervals of radii $M - r_j$ have Minkowski sum of radius

$$\sum_j (M - r_j) = nM - R \geq |c|.$$

Now set

$$G_j = F_j - c_j + d_j.$$

Then $\sum_j G_j(\beta_j x) = f(x)$ and $\|G_j\|_\infty = r_j + |d_j| \leq M$.

For sharpness take $f(x) = \cos(2\pi\beta_1 x)$. The nonzero Fourier spectra of the periodic summand spaces are the disjoint lattices $\beta_j \mathbb{Z} \setminus \{0\}$. Uniqueness of Fourier coefficients forces the first summand in any decomposition to be $\cos(2\pi\beta_1 x)$ plus a constant, so its norm is at least one. $\square$

4 The rank-two three-period argument

We first handle the genuinely new case in which the three periods are pairwise incommensurable. Take an initial decomposition and let

$$K = \overline{\{(\beta_1 x, \beta_2 x, \beta_3 x) \bmod 1 : x \in \mathbb{R}\}} \subset \mathbb{T}^3.$$

Every two-coordinate projection of K is onto $\mathbb{T}^2$. Let μ be Haar measure on K , put $m_j = \int_{\mathbb{T}} F_j$, and $m = \sum_j m_j = \int_K f$. Define

$$h_j = F_j + m - m_j.$$

The pullback of h_j to K is the conditional Haar average of f over the fibres of the j th coordinate: the other two coordinates are Haar-uniform because the corresponding two-coordinate projections are onto. Therefore, with $M = \|f\|_\infty$,

$$-M \leq h_j \leq M, \quad \int_{\mathbb{T}} h_j = m, \quad h_1 + h_2 + h_3 = f + 2m. \quad (1)$$

Write $\ell_j = \min h_j$ and $u_j = \max h_j$. If $m \geq 0$, choose a point of K at which any prescribed pair h_i, h_j attains its two minima; this is possible by surjectivity of the (i, j) coordinate projection. At that point $f \geq -M$, while the remaining $h_k \leq M$. From (1),

$$\ell_i + \ell_j \geq 2m - 2M.$$

Adding the three pair inequalities yields

$$\sum_{j=1}^3 (\ell_j + M) \geq 3m \geq 2m.$$

Thus there are $q_j \in [0, \ell_j + M]$ with $\sum_j q_j = 2m$. The functions $g_j = h_j - q_j$ sum to f and remain in $[-M, M]$.

If $m \leq 0$, apply the symmetric argument at simultaneous pairwise maxima. It gives

$$u_i + u_j \leq 2m + 2M, \quad \sum_{j=1}^3 (M - u_j) \geq -3m \geq -2m.$$

Choose $q_j \in [0, M - u_j]$ with $\sum_j q_j = -2m$ and put $g_j = h_j + q_j$. Again the summands add to f and have norm at most M .

Proof of Theorem 2. The pairwise incommensurable case is proved above. Otherwise exactly two periods, say α_1, α_2 , are commensurable and the third is incommensurable with both. The sum of the first two initial components has a common period L . Apply Theorem 1 for two periods to this aggregate and the α_3 -periodic component. Its construction only moves a constant, so it produces

$$f = H + g_3, \quad \|H\|_\infty, \|g_3\|_\infty \leq M,$$

with $H \in \mathcal{P}_{\alpha_1} + \mathcal{P}_{\alpha_2}$. The source's sharp two-period result decomposes $H = g_1 + g_2$ with $\|g_1\|_\infty, \|g_2\|_\infty \leq \|H\|_\infty \leq M$.

Sharpness follows as in Theorem 1, using a cosine whose frequency belongs to the singleton commensurability class in the second case. $\square$

5 Boundary of the result

The only three-period case not covered is the fully commensurable one. It reduces fibrewise to a norm-one splitting problem for three coset partitions of a finite cyclic group. Exhaustive small vertex computations and broader sampled searches found quotient norm one, but no uniform dual certificate or continuous selection argument was obtained. For four or more distinct commensurability classes, higher rational relations create the analogous proper-subtorus obstruction.

No external record settling the exact constant question was located in the bounded search performed for this packet. The local parsed corpus contains the question only in arXiv:1312.3798; the adjacent arXiv:1401.1226 develops semigroup decomposition results but does not settle this norm-one problem.

Human-review recommendation. Check the fibre conditional-average identity, the pair-extrema inequalities, and the reduction with one commensurable pair.

References

- [1] B. Farkas and Sz. Gy. Révész, *The periodic decomposition problem*, arXiv:1312.3798v1 (2013).